

Student: Jonatan Rabjerg
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Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 6

Consider the series $\sum_{n=0}^{\infty} (7x)^n$

- (a) Find the series' radius and interval of convergence.
 (b) For what values of x does the series converge absolutely?
 (c) For what values of x does the series converge conditionally?

(a) To find the radius and interval of convergence, first notice that this series is a geometric series.

Equate the terms of the geometric series $\sum_{n=0}^{\infty} ar^n$ to the given series $\sum_{n=0}^{\infty} (7x)^n$ to find expressions for a and r in terms of x .

$$a = 1 \text{ and } r = 7x$$

The geometric series $\sum_{n=0}^{\infty} ar^n$ converges if $|r| < 1$ and diverges if $|r| \geq 1$.

So, the value of a does not affect the interval of convergence. To find the interval of convergence, use the convergence condition for the geometric series and substitute $7x$ for r .

$$\begin{aligned} |r| &< 1 \\ |7x| &< 1 \end{aligned}$$

Replace the absolute value inequality with a compound inequality and solve for x .

$$\begin{aligned} -1 &< 7x < 1 && \text{Rewrite the absolute value inequality.} \\ -\frac{1}{7} &< x < \frac{1}{7} && \text{Divide all terms by 7.} \end{aligned}$$

Therefore the interval of convergence is $\left(-\frac{1}{7}, \frac{1}{7}\right)$.

Now find the radius of convergence, R . The Corollary to the Convergence Theorem for Power Series states that the convergence of a power series centered at $x = a$, $\sum_{n=0}^{\infty} c_n(x - a)^n$, is described by one of the following three cases.

1. There is a positive number R such that the series diverges for x with $|x - a| > R$ but converges absolutely for x with $|x - a| < R$.
2. The series converges absolutely for every x , in which case the radius of convergence is infinity.
3. The series converges at $x = a$ and diverges elsewhere, in which case R is equal to zero.

Since the interval of convergence for this series has a finite and positive width, it must be the first case.

Since the interval of convergence is both finite and positive, the interval is centered at a , with a radius of convergence equal to $|x - a|$, or half the width of the interval. Find half the width of the interval to find the radius.

$$\begin{aligned} R &= \frac{1}{2} \left(\frac{1}{7} - \left(-\frac{1}{7} \right) \right) \\ &= \frac{1}{7} \end{aligned}$$

(b) Now determine where the series converges absolutely. First note from the Corollary to the Convergence Theorem that any power series with a finite and positive radius of convergence converges absolutely for any x within the open interval of convergence. The only other points that need to be checked are the endpoints, $-\frac{1}{7}$ and $\frac{1}{7}$, where the series may converge even if it does not converge absolutely.

Here the geometric series can still be used to determine whether the series converges absolutely at the endpoints. At either endpoint $x = -\frac{1}{7}$ or $x = \frac{1}{7}$, $|7x| = |r| = 1$, which means the series does not converge since the geometric series diverges for $|r| \geq 1$.

Therefore, the series $\sum_{n=0}^{\infty} (7x)^n$ converges absolutely on the same interval that it converges, or $\left(-\frac{1}{7}, \frac{1}{7}\right)$.

(c) Now find the values of x for which the series converges conditionally. A series is called conditionally convergent if it is convergent but is not absolutely convergent.

Since the series converges for all x in the interval $\left(-\frac{1}{7}, \frac{1}{7}\right)$ and converges absolutely for all x in the interval $\left(-\frac{1}{7}, \frac{1}{7}\right)$, it does not converge conditionally for any value of x .